# CardioAI Model Evaluation Results

Final comparison of evaluated machine learning models for cardiovascular risk assessment.

| Model | Accuracy | Precision | Recall | F1-score | ROC-AUC |
|---|---|---|---|---|---|
| KNN | 0.9022 | 0.8889 | 0.9412 | 0.9143 | 0.937590 |
| Logistic Regression | 0.891304 | 0.894231 | 0.911765 | 0.902913 | 0.934840 |
| Random Forest | 0.880435 | 0.857143 | 0.941176 | 0.897196 | 0.929340 |
| SVM | 0.869565 | 0.882353 | 0.882353 | 0.882353 | 0.937351 |

**Selected model:** K-Nearest Neighbors (KNN). The KNN row above was updated using the calibrated KNN evaluation values after probability calibration.

### Calibrated KNN metrics

After calibration, KNN achieved Accuracy = **0.9022**, Precision = **0.8889**, Recall = **0.9412**, and F1-score = **0.9143**.

### Calibration note

After selecting KNN as the best model, probability calibration was applied to improve the reliability of predicted probabilities. As a result, the final decision threshold changed from 0.43 before calibration to 0.35 after calibration. Therefore, the final confusion matrix is reported using the calibrated KNN threshold of 0.35.